

Zhenwei Dai

📍 Seattle ✉ daizwhao@gmail.com ☎ +1 408-581-3594 🌐 Personal Website in LinkedIn Profile

Professional Summary

Principal Applied Scientist at Microsoft M365 Copilot, leading LLM post-training and evaluation for production agents. Four years of industry experience across Microsoft and Amazon in supervised fine-tuning, reinforcement learning, distillation, tool use, and search. Recent research includes [MiniCorp](#), an office simulator for AI agents, and publications at NeurIPS, ACL, ICLR, and SIGIR.

Education

Rice University
PhD in Statistics

Sept 2018 – Aug 2022

- **Advisors:** Anshumali Shrivastava and Reinhard Heckel
- **Research Directions:** Large Scale machine learning efficiency, Randomized algorithms, Data mining

Experience

Principal Applied Scientist *Microsoft M365 Copilot*

Redmond, WA
Dec 2025 – Present

- Led end-to-end post-training of GPT-5 series models for M365 Copilot using OpenAI's LLM training infrastructure, covering the full pipeline: SFT, reinforcement learning (RL), and on-policy distillation to improve the Copilot response quality and agent inference efficiency.
- Developed rigorous offline and online evaluation benchmarks for M365 Copilot features, enabling systematic measurement of post-training gains and validating quality improvements at production scale.

Senior Applied Scientist *Amazon Ads (LLM Agent)*

Seattle, WA
May 2025 – Nov 2025

- Led the science strategy and infrastructure development for a large-scale LLM agent designed to proactively assist Amazon advertisers with practical challenges, such as setting up advertising campaigns, optimizing budgets, and navigating business concerns like tariff impacts.
- Built a scalable post-training platform on Amazon EKS, orchestrating SFT and reinforcement learning (RL) pipelines across a distributed cluster (around 100 machines and 1000 H200 GPUs) to effectively teach the agent to leverage Amazon Ads-specific tools.
- Defined and led the post-training roadmap for a tool-augmented LLM agent, enabling it to learn and use Amazon Ads tools to address advertisers' questions and recommend actionable strategies.

Senior Applied Scientist *Amazon Search*

Palo Alto, CA
May 2023 – April 2025

- Collaborated cross-team on Amazon Rufus applications, leading post-training (primarily prompt engineering) efforts to effectively adapt instructed LLMs for reliable responses and rapid customization to specific product scenarios [8, 7].
- Developed and deployed transformer-based query understanding models for large-scale search intent identification within Amazon Search. These models can determine the customers' search intent and bridge the search query with Amazon product catalog [12].
- Actively mentored research interns, leading to multiple collaborative publications at top-tier NLP and LLM venues (such as ACL and ICML), while providing science guidance and supporting intern career development.

Applied Scientist *AWS AI Research and Education (AIRE)*

Santa Clara, CA
Sept 2022 – April 2023

- Led the development of an LLM-based synthetic benchmark for discovering relations between tabular datasets, enabling evaluation without access to real customer data.
- Developed machine learning algorithms for discovering relationships among tabular datasets (e.g., joinability and unionability) in AWS data lakes.

Selected Publications

- [1] Jingying Zeng, **Zhenwei Dai**, Jinning Li, Changho Shin, Dylan Zhang, Yuxuan Lu, Qi He, Dakuo Wang, Kai-Wei Chang. “MiniCorp: The Last Mile of the AI Agent Firm”. In: *arXiv preprint arXiv:2610.05912* (2026). URL: <https://agents-minicorp.com/>.
- [2] Yuxuan Lu, Ziyi Wang, Yingzhou Lu, Yisi Sang, Jiri Gesi, Xianfeng Tang, Yimeng Zhang, **Zhenwei Dai**. “Firefly: Illuminating Verified Real-world Tool Call Data Generation”. In: *NeurIPS 2026 (poster)* (2026).
- [3] Pengfei He, **Zhenwei Dai**, Xianfeng Tang, Yue Xing, Hui Liu, Jingying Zeng, Qiankun Peng, Shrivats Agrawal. “To Trust or Not to Trust: Attention-based Trust Management for LLM Multi-Agent Systems”. In: *ACL 2026 (main conference); NeurIPS 2026 (poster)* (2026).
- [4] Yingqian Cui, **Zhenwei Dai**, Pengfei He, Bing He, Hui Liu, Zhan Shi, Xianfeng Tang, Jingying Zeng. “A Reward-Guided Dual-Phase Framework for Adaptive Inference-Time Reasoning”. In: *Findings of ACL 2026* (2026).
- [5] Pengfei He, **Zhenwei Dai**, Bing He, Hui Liu, Xianfeng Tang, Hanqing Lu, Juanhui Li, Jiayuan Ding, Subhabrata Mukherjee, Suhang Wang. “TRAJECT-Bench: A Trajectory-Aware Benchmark for Evaluating Agentic Tool Use”. In: *ICLR* (2026).
- [6] Jie Ren, **Zhenwei Dai**, Xianfeng Tang, Yue Xing, Shenglai Zeng, Hui Liu, Jingying Zeng, Qiankun Peng, Samarth Varshney, Suhang Wang. “Keeping an eye on llm unlearning: The hidden risk and remedy”. In: *NeurIPS* (2025).
- [7] Jingying Zeng*, **Zhenwei Dai***, Hui Liu*, Samarth Varshney, Zhiji Liu, Chen Luo, Zhen Li, Qi He, Xianfeng Tang. “Examples as the Prompt: A Scalable Approach for Efficient LLM Adaptation in E-Commerce”. In: *SIGIR industry track* (2025).
- [8] Jingying Zeng, **Zhenwei Dai***, Hui Liu, Xianfeng Tang, Chen Luo, Samarth Varshney, Zhen Li, Qi He. “Cite before you speak: Enhancing context-response grounding in e-commerce conversational llm-agents”. In: *arXiv preprint arXiv:2503.04830* (2025).
- [9] Jie Ren*, **Zhenwei Dai***, Xianfeng Tang, Hui Liu, Jingying Zeng, Zhen Li, Rahul Goutam, Suhang Wang, Yue Xing, Qi He. “A General Framework to Enhance Fine-tuning-based LLM Unlearning”. In: *ACL findings* (2025).
- [10] Yingqian Cui, Pengfei He, Jingying Zeng, Hui Liu, Xianfeng Tang, **Zhenwei Dai**, Yan Han, Chen Luo, Jing Huang, Zhen Li. “Stepwise perplexity-guided refinement for efficient chain-of-thought reasoning in large language models”. In: *ACL findings* (2025).
- [11] Minhua Lin, Hui Liu, Xianfeng Tang, Jingying Zeng, **Zhenwei Dai**, Chen Luo, Zheng Li, Xiang Zhang, Qi He, Suhang Wang. “How Far are LLMs from Real Search? A Comprehensive Study on Efficiency, Completeness, and Inherent Capabilities”. In: *arXiv preprint arXiv:2502.18387* (2025).
- [12] **Zhenwei Dai**, Chen Luo, Zhen Li, Xianfeng Tang, Hanqing Lu, Rahul Goutam, Haiyang Zhang. “RA-NER: Retrieval augmented NER for knowledge intensive named entity recognition”. In: *The Second Tiny Papers Track at ICLR 2024*. 2024.
- [13] **Zhenwei Dai**, Vasileios Ioannidis, Soji Adeshina, Zak Jost, Christos Faloutsos, George Karypis. “ScatterSample: Diversified Label Sampling for Data Efficient Graph Neural Network Learning”. In: *Learning on Graphs Conference* (2022).
- [14] **Zhenwei Dai**, Aditya Desai, Reinhard Heckel, Anshumali Shrivastava. “Active Sampling Count Sketch (ASCS) for Online Sparse Estimation of a Trillion Scale Covariance Matrix”. In: *Proceedings of the 2021 ACM SIGMOD International Conference on Management of Data*. 2021.
- [15] **Zhenwei Dai**, Anshumali Shrivastava. “Adaptive Learned Bloom Filter (Ada-BF): Efficient Utilization of the Classifier with Application to Real-Time Information Filtering on the Web”. In: *Advances in Neural Information Processing Systems*. Vol. 33. 2020, pp. 11700–11710.
- [16] **Zhenwei Dai**, Reinhard Heckel. “Channel Normalization in Convolutional Neural Network Avoids Vanishing Gradients”. In: *ICML Workshop Deep Phenomena* (2019).